OSI layers
(Open System Interconnection)

1. Physical layer: topology used, mechanical & electrical specifications.
2. Data link layer: error and flow control, physical adds. (MAC adds.)
   Hop-to-hop delivery; error & flow control done by trailers.
3. Network layer: delivery of individual pkts to final destination
   logical addressing (IP address), routing the pkts.
4. Transport layer: service pt. or port addressing, error & flow control
   Segmentation and reassembly
   Process-to-process delivery
5. Session layer: establish, maintain, synchronize interaction b/w systems
   adds checkpts. (dialog control and synchronisation)
6. Presentation layer: syntax and semantics
   data compression, encryption, decryption
7. Application layer: protocols to be used, remote login, accessing www

TCP / IP Protocol Suite

1. Physical layer	Physical adds.	Bits
2. Data link layer		Frames
3. Network layer	Logical adds.	Packets
4. Transport layer	Port adds.	
5. Application layer	Specific adds.	Data stream

Addresses

- Physical
- Logical
- Port
- Specific

Network Devices

- Modem modulates digital signal to analog signal
- Repeater: regenerate the signal

1. Hub - a distributor having many ports connected to computers
2. Switch - transmit pkg. to destination
3. Router
4. Gateway

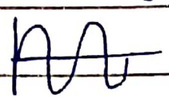
OSI	TCP/IP
1. 7 layers	1. 4 layers
2. less reliable	2. more reliable
3. general model	3. cannot use any other application hence not a general model
4. follows HORIZONTAL APPROACH	4. VERTICAL APPROACH
5. Separate presentation layer	5. no separate presentation layer
6. transport layer guarantee delivery of data	6. doesn't guarantee
7. OSI has problem of fitting the protocol in the model	7. doesn't fit any protocol
8. <u>N/W layer:</u> connection-oriented and connectionless	8. <u>N/W layer:</u> connectionless
9. <u>Transport layer:</u> connection oriented	9. <u>Transport layer:</u> connectionless and connection-oriented
10. provides separate SERVICE (what the layer does), INTERFACE (how to access service), PROTOCOL (how to perform service)	10. doesn't
11. Protocols are hidden in OSI and easily replaced as technology changed	11. Replacing protocol not easy and protocols are not hidden

PHYSICAL LAYER:

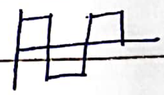
1. Representation of bits
2. Topology
3. Synchronisation (b/w transmitter and receiver)
4. line configuration (like pt.-to-pt., multipt.)
5. Transmission mode (eg. half duplex, full duplex)
6. Data rate
7. Interface (transmission medium)

Classification of signals

→ Analog



→ Digital



SIGNALS
- Periodic
- Aperiodic

Analog Signal

→ Simple
eg. Sine wave

→ Composite
eg. multiple sine wave

HARMONICS - signal multiple frequency

SINE WAVE - Simple analog signal

Sine wave is periodic.

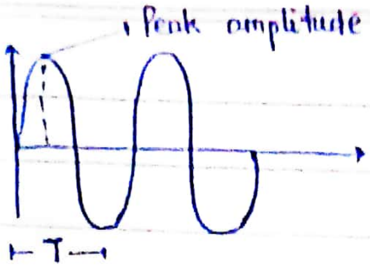
$$s(t) = A \sin((2\pi f t) + \theta)$$

A → Peak value of amplitude

f → freq.

θ → Phase

1. Peak amplitude :



2. Period (T) and frequency (f) :

T → amt. of time to complete 1 cycle

UNIT : seconds

f → no. of periods in one cycle

$$f = \frac{1}{T}$$

3. Phase : 1 rad = $\frac{360}{2\pi}$

↳ • Position of waveform selected to zero

• Status of cycle

UNIT : degree / radian

Digital Signal

1. BIT LENGTH : distance one bit occupies in transmission medium

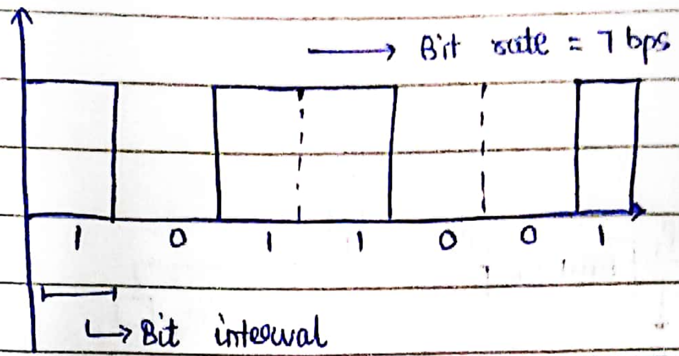
2. BIT INTERVAL : time taken to send one single bit

3. BIT RATE : transmission speed of bits

no. of bits transmitted in per unit time

Unit - bps (bit / second)

4. BAUD RATE : → No. of signals units per second
→ Data required to represent these bits
→ Also called Signalling or modulation rate
→ Unit : Baud (Bd) or signal / sec
→ Indicates rate at which signal level changes over given period of time



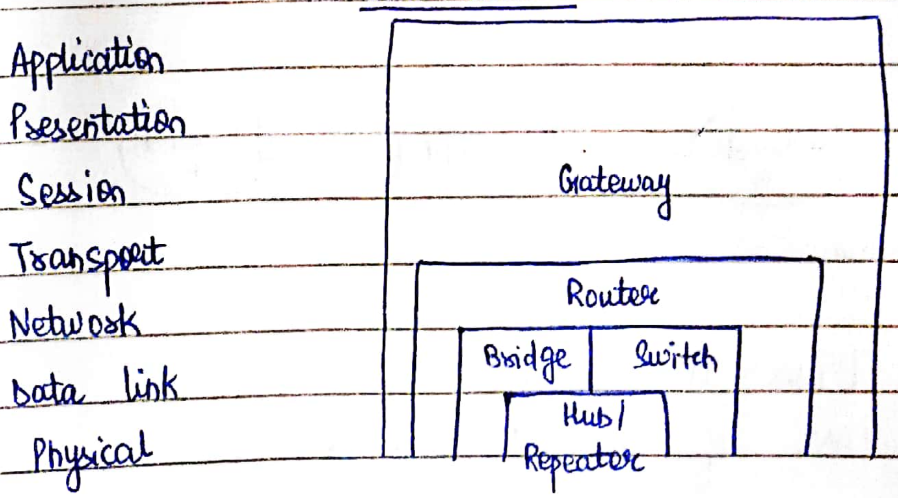
$$\boxed{\text{Bit rate} = \text{Baud rate} \times \text{No. of bits / sec.}}$$

Q. Analog signal carries 4 bits in each signal unit. If 1000 signals units are sent per sec., find the baud rate and bit rate.

Sol.ⁿ Baud rate = 1000 Bd

Bit rate = $1000 \times 4 = 4000$ bps

N/W devices

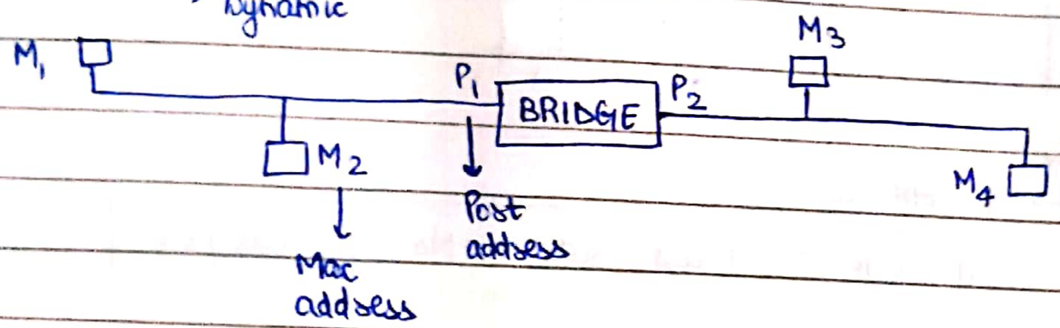


Repeater is 2-port while hub is multiport.

<u>HUB</u> : It broadcasts the data.

- → security link
- → forwarding
- → collision domain
- → multiport
- → connects similar LANs
- → uses half duplex method

<u>BRIDGES</u> : → Static / → Dynamic



Bridge works as per the mac addresses of the sender and receiver

- → store and forward
- → filtering
- → less collision than hub
- → connects 2 dissimilar LANs
- → 2-port

MAC ADDR.	PORT

<u>ROUTERS</u> :

- → collision domain
- → forwarding
- → filtering (uses IP address).
- → connects LANs of different networks

DEVICE	~~APPLICATION~~ LAYER	DEAL	ADDRESS
1. Gateway	Application	Msg.	IP addr.
2. Gateway	Presentation ~~Physical~~	"	"
3. Gateway	Session	"	"
4. Gateway	Transport	Pkt. / Msg.	"
5. Gateway + Router	Network	Pkt.	"
6. Gateway + Router + Bridge / Switch	Data link	Frames	MAC addr.
7. Hub / Repeater + Gateway + Router + Bridge	Physical	Bits / Signals	Broadcast or one-to-one port

Q. Bit rate = 3000 bps
Each signal unit carries 6 bits. Find the baud rate

Sol. Baud rate = $\frac{3000}{6} = 500$ Bd

Q. An analog signal has a bit rate of 8000 bps and baud rate 1000 Baud. How many data elements are carried by each signal element?
How many signals do we need?

Sol. No. of bits in each signal, $n = \log_2 L$

$n = \frac{8000}{1000} = 8$

$\boxed{L = 2^n}$
↓
Signal element
↓
smallest unit of signal

$L = 2^n = 2^8 = 256$

Q. Assume we need to download text doc. @ 100 pg/min. What is the required bit rate of the channel?

Sol.ⁿ

$1 \text{ pg} = 24 \text{ lines}$

$1 \text{ line} = 80 \text{ characters}$

$1 \text{ char} = 8 \text{ bits}$

No. of bits $= 100 \times 24 \times 80 \times 8$

$= 1536000$ bits

Bit rate $= \dfrac{153600\cancel{0}}{6\cancel{0}} = 25600$

Q. A digital signal has 8 levels (signals). How many bits are needed in one level?

Sol.ⁿ

$L = 2^n$

$8 = 2^n$

$n = 3$

CIRCUIT SWITCHING	PACKET SWITCHING
→ Physical layer	→ Network layer
→ Dedicated path	→ Each msg. is treated as a packet and is treated independently.
→ Contiguous flow	→ No setup or teardown phase
→ No out of order	→ 1. Datagram - NL 2. Virtual circuit - DL
→ No headers required	→ Store and forward
Steps: 1. Setup 2. Transmission 3. Tear down	→ High efficiency
→ less efficiency and minimal delay due to reserved bandwidth	→ more delay
	→ Pipelining

- Total time = Setup time + Transmission time + Propagation time + Tear down time

- Total time = n(Transmission time) + Propagation time

- Propagation time = $\frac{\text{distance}}{\text{speed}}$

- Transmission time = $\frac{\text{message}}{\text{bandwidth}}$

DATAGRAM	VIRTUAL CIRCUIT
→ Connectionless	Connection oriented
→ Network layer	Data link layer
→ no reservation	reservation
→ high overhead	less overhead
→ Packet lost high	less
→ used in internet	Frame relay, atm
→ more delay	less delay

Message switching - same as pkt. switching, the only difference is message is sent as a whole, not ~~decided~~ divided into pkts.

UNIT - 2

Types of errors

1. Bit errors - one bit changes
2. Burst errors - 2 or more bits change

ERROR DETECTION

- Parity check
  - VRC (Vertical Redundancy check)
  - LRC (Longitudinal check)
- CRC Cyclic Redundancy Check
- CHECKSUM

101011 ⟶ 111111   length of Burst error = 3

Dataword	Codeword
101	101 01
k	↳ code r

$n = k + r = 3 + 2 = 5$

CRC (Cyclic Redundancy Check)

Divisor: 1101 $0 - 1 = 1$

Data: 100100 $0 - 0 = 0$

$1 - 0 = 1$

Add (no. of bits in divisor - 1) 0's to data.

∴ Data becomes 100100000

```
1101 ) 100100000 (
```

Keep dividing.

If 2 consecutive 0's appear on the left, next bit of quotient will be 0.

Codeword = Data + Remainder

Divide by divisor. If rem. is 0 there's no error.

Q. The codeword is received as 1100 1001 01011. Check if there exists an error if divisor is 10101.

Sol.ⁿ

```
10101 ) 1100100101011 ( 111110001
        10101
        -----
         11000
         10101
         -----
          11010
          10101
          -----
           11111
           10101
           -----
            10100
            10101
            -----
                11
                00
               ---
                110
                000
               ----
                1101
                11011
                10101
               -----
                 1110  → not 0 so there's an error.
```

Q. Data: 110101011

Generator: $x^4 + x + 1$

Sol^n

```
10011 ) 1101010110000 ( 1100000011
        10011↓
         10011
         10011↓↓↓↓↓
              01100      11000
              10011      10011
               11110      10110
               10011      10011
                11010      0101  ⇒ x^2+1
                10011
                 1001
```

Codeword = 110101011 + 0101

Q. → $x^7 + x^5 + 1$

→ $x^3 + 1$

Sol^n

```
1001 ) 10100001000 ( 10110111
       1001↓↓
         1100
         1001↓
          1010
          1001↓↓
            1110
            1001↓
             1110
             1001
              1110
              1001
               111  → x^2+x+1
```

Codeword = 10100001 + 111

CHECKSUM

(7, 12, 5, 9)

(7, 12, 5, 9, -33)

↳ checksum

Hamming code

No. of information bits	No. of parity bits
2 - 4	3
5 - 11	4
12 - 26	5
27 - 57	6
58 - 120	

Parity	Bits to be checked
P_1	1, 3, 5, 7, 9, 11, 13
P_2	2, 3, 6, 7, 10, 11, 14, 15
P_4	4, 5, 6, 7, 12, 13, 14, 15
P_8	8, 9, 10, 11, 12, 13, 14, 15

Correction method

		PARITY
0	0000	
1	0001	P_1
2	0010	P_2
3	0011	P_1, P_2
4	0100	P_4
5	0101	P_1, P_4
6	0110	P_2, P_4
7	0111	P_1, P_2, P_4
8	1000	P_8

	P_{16} P_8 P_4 P_2 P_1	PARITY
9	1001	P_1, P_8
10	1010	P_8, P_2
11	1011	P_8, P_2, P_1
12	1100	P_8, P_4
13	1101	P_8, P_4, P_1
14	1110	P_8, P_4, P_2
15	1111	P_8, P_4, P_2, P_1
16	10000	P_{16}
17	10001	P_{16}, P_1
18	10010	P_{16}, P_2

Q. Bit word : 1011

Construct even parity.

Seven bit hamming code.

Sol:- Data bits : 1011

Parity bits = 3 (P_1, P_2, P_4)

1	0	1		1		
D_7	D_6	D_5	P_3	D_3	P_2	P_1

(i) $P_1 \rightarrow$ check 1,3,5,7

$\rightarrow$ 1,1,1

$P_1 = 1$ (for even parity)

(ii) $P_2 \rightarrow$ check 2,3,6,7

$\rightarrow$ 101

$P_2 = 0$

(iii) $P_4 \rightarrow$ check 4,5,6,7

$\rightarrow$ 101

$P_4 = 0$

1	0	1	0	0	0	1
D_7	D_6	D_5	P_4	D_3	P_2	P_1

Receiver end: suppose D_3 is toggled and data becomes 1010101.

$\bar{D_3}$

P_1 : 1,3,5,7

= 1,0,1,1 $\rightarrow$ odd

$P_1 = 1$ (error)

P_2 : 2,3,6,7

= 0,0,0,1 $\rightarrow$ odd

$P_2 = 1$ (error)

P_4 : 4,5,6,7

= 0,1,0,1 $\rightarrow$ even

$P_4 = 0$

Q. A 7-bit hamming code is received as 1110101. What will be the correct code?

Sol.ⁿ

1	1	1	0	1	0	1
D_7	D_6	D_5	P_4	D_3	P_2	P_1

$P_1 : 1,3,5,7$
$= 1,1,1,1$
$P_1 = 0$

$P_2 : 2,3,6,7$
$= 0,1,1,1$
$P_2 = 1$

$P_4 : 4,5,6,7$
$= 0,1,1,1$
$P_4 = 1$

$P_4\ P_2\ P_1$
$1\ 1\ 0 = 6$

D_6 has an error.

Q.1. Write the steps to generate the hamming code. Prepare a code for bit pattern (1010). Suppose while transmitting, error occurs at 7th bit. Write the bit pattern for receiver.

Q.2. Encode the following 8-bit data using hamming code.
(a) 10101111 (even parity)
(b) 01111100 (odd parity)

Sol.1.)

D_7	D_6	D_5	P_4	D_3	P_2	P_1
1	0	1	0	0	1	0

$P_1 \rightarrow 0,1,1$
$P_2 \rightarrow 0,0,1$
$P_4 \rightarrow 1,0,1$

Codeword is 1010010.

But there's an error at 7th bit, so receiver will receive the data as 0010010.

Sol 2.) (a)

1	0	1	0	0	1	1	1	0	1	0	1
D_{12}	D_{11}	D_{10}	D_9	P_8	D_7	D_6	D_5	P_4	D_3	P_2	P_1

P_1 : 1,1,1,0,0
$P_1 = 1$

P_2 : 1,1,1,1,0
$P_2 = 0$

P_4 : 1,1,1,1
$P_4 = 0$

P_8 : 0,1,0,1
$P_8 = 0$

The encoded data will be 101001 110101.

(b)

0	1	1	1	0	1	1	0	1	0	1	0
D_{12}	D_{11}	D_{10}	D_9	P_8	D_7	D_6	D_5	P_4	D_3	P_2	P_1

P_1 : 0,0,1,1,1

P_2 : 0,1,1,1,1

P_4 : 0,1,1,0

P_8 : 1,1,1,0

The encoded data will be ~~011101 100001~~ $\boxed{011101\ 101010}$

Q. A receiver receives the following hamming code with odd parity : 001111001011101. Find the error bit.

Sol.ⁿ

0	0	1	1	1	1	0	0	1	0	1	1	1	0	0	1	1	0
D_{18}	D_{17}	P_{16}	D_{15}	D_{14}	D_{13}	D_{12}	D_{11}	D_{10}	D_9	P_8	D_7	D_6	D_5	P_4	D_3	P_2	P_1

P_1 : 1,3,5,7,9,11,13
1,0,1,0,0,1
$P_1 = 0$

P_2 : 2,3,6,7,10,11,14,15
1,1,1,1,0,1,1
$P_2 = 1$

P_4 : 4,5,6,7,12,13,14,15
0,1,1,0,1,1,1
$P_4 = 0$

P_8 : 8,9,10,11,12,13,14,15
0,1,0,0,1,1,1
$P_8 = 1$

$P_{16} = 1$

16 8 4 2 1
1 1 0 1 0

FLOW CONTROL

Stop & wait

- → only 1 frame
- → sender window = 1
- receiver window = 1

→ $\eta = \dfrac{T_t}{T_t + 2T_p}$

Go-Back N

- multiple frames
- sender window = $2^{K}-1$
- receiver window = 1

$\eta = (2^K - 1) \cdot \dfrac{T_t}{T_t + 2T_p}$

Selective repeat

- multiple frames
- sender window = $2^{K}-1$
- receiver window = $2^{K}-1$

$\eta = (2^K - 1) \cdot \dfrac{T_t}{T_t + 2T_p}$

1. TRANSMISSION TIME = $T_t = \dfrac{L}{BW}$

2. TOTAL TIME = $T_t + T_p$ + Queuing + Processing = $T_t + 2T_p$

3. $T_p = \dfrac{d}{r}$

CRC Questions

Q.1. The codeword is received as 1100 1001 01011. Check whether there are errors in the received codeword, if the divisor is 10101 (The divisor corresponds to the generator polynomial).

Sol.ⁿ

```
10101 ) 1100100101011 ( 111110001
        10101
        -----
         11000
         10101
         -----
          11010
          10101
          -----
           11110
           10101
           -----
            10111
            10101
            -----
              10011
              10101
              -----
               1110
```

Since the remainder is not 0, hence there's an error.

Q.2) Generate the CRC code for the data word of 110010101. The divisor is 10101.

Sol.ⁿ

```
10101 ) 1100101010000 ( 111110111
        10101
        -----
         11000
         10101
         -----
          11011
          10101
          -----
           11100
           10101
           -----
            10011
            10101
            -----
            011000
              10101
              -----
               11010
               10101
               -----
                11110
                10101  →  1011
```

Codeword = Data + Remainder

= 110010101 + 1011●

= 1100101010110 11●

Q.3) Write the steps to compute the checksum in CRC code. Calculate CRC for the frame 110101011 and the generator polynomial $= x^4 + x + 1$ and write the transmitted frame.

Sol.ⁿ Divisor = $\overset{4}{1}\overset{3}{0}\overset{2}{0}\overset{1}{1}\overset{0}{1}$

```
10011 ) 1101010110000 ( 11000001
        10011
        -----
         10011
         10011
         -----
            0011000
            ~~10011~~
            -------
            ~~10101~~
              10011
              -----
               10110
               10011
               -----
                 101
```

Codeword = Data + Remainder

= 110101011 + 101

= 110101011101

Q.4) If the frame is 110101011 and generator is $x^3 + x + 1$. What would be the transmitted frame?

Sol.ⁿ Divisor = $\overset{3}{1}\overset{2}{0}\overset{1}{1}\overset{0}{1}$

⇒ 110101011 + 000

1011) 11010101011000 (111101101

1011
1100
1011
1111
1011
1000
1011
1111
1011
1000
1011
1100
1011
111

Transmitted form = 110101011 + 111

= 110101011111

5.) What is the remainder obtained by dividing x^7+x^5+1 by the generator polynomial x^3+1?

Sol.ⁿ Data is

7	6	5	4	3	2	1	0
1	0	1	0	0	0	0	1

Divisor is

3	2	1	0
1	0	0	1

1	0	1	0	0	0	0	1	0	0	0
10	9	8	7	6	5	4	3	2	1	0

$x^3+1 \,)\, x^{10}+x^8+x^3 \,(\, x^7+x^5+x^4+x^2+x$

$\cancel{x^{10}}+x^7$

$\cancel{x^8}+x^7+x^3$

$\cancel{x^8}+x^5$

$\cancel{x^7}+x^5+x^3$

$\cancel{x^7}+x^4$

$\cancel{x^5}+x^4+x^3$

$\cancel{x^5}+x^2$

$\cancel{x^4}+x^2+x^3$

x^4+x

$$
\begin{array}{r|l}
x^3+1 & x^3+x^2+x \;(\,1 \\
 & \underline{x^3 \qquad\;\; +1} \\
 & \quad x^2+x+1 \rightarrow \text{remainder}
\end{array}
$$

Remainder obtained is 111.

Q.6) Calculate CRC for a 10 bit sequence 1010011110. The generator polynomial is x^3+x+1.

Sol.ⁿ Divisor = 1011 (bit positions 3 2 1 0)

```
1011 ) 1010011110000 ( 1001000111
       1011
       ----
          1011
          1011
          ----
           01100
            1011
            ----
             1110
             1011
             ----
              1010
              1011
              ----
               001
```

CRC = 1010011110 + 001

= 1010011110001

Q.7) A bit stream 10011101 is transmitted using the standard CRC method. The generator polynomial is x^3+1. Show the actual bit string transmitted. Suppose the third bit from left is inverted during transmission. Show that this error is detected at the receiver end.

Sol^n Divisor = $\begin{array}{cccc} 3 & 2 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{array}$

```
1001 | 10011101000 | 10001100
       1001
        1101
        1001
         1000
         1001
          0 100
```

Codeword = Data + Remainder
= 10011101000 + 100
= 10011101000100

```
1001 | 10111101000100 | 1010 1000 DO
       1001
        1011
        1001
          1001
          1001
           0001 100  → Remainder
           ~~10001~~
           ~~1000~~
```

Remainder is 100 hence there's an error.

Hamming Code Questions

1. If the 7-bit Hamming codeword received by a receiver is 1011011. Assuming the even parity, state whether the received codeword is correct or wrong. If wrong, locate the bit in error.

Sol.ⁿ

D_7	D_6	D_5	P_4	D_3	P_2	P_1
1	0	1	1	0	1	1

(i) P_1: check 1, 3, 5, 7

→ 1, 0, 1, 1

$P_1 = 0$

(ii) P_2: check 2, 3, 6, 7

→ 1, 0, 0, 1

$P_2 = 1$

(iii) P_4: check 4, 5, 6, 7

→ 1, 1, 0, 1

$P_4 = 0$

The correct codeword will be 1010011.

P_4 P_2 P_1

0 1 0 ⇒ II bit has error.

The 2nd bit should be 0 and not 1.

2. A seven bit hamming code is received as 1110101. What will be the correct code?

D_7	D_6	D_5	P_4	D_3	P_2	P_1
1	1	1	0	1	0	1

Sol.ⁿ P_1: 1, 3, 5, 7

1, 1, 1, 1

$P_1 = 0$

P_2: 2, 3, 6, 7

0, 1, 1, 1

$P_2 = 1$

P_4: 4, 5, 6, 7

0, 1, 1, 1

$P_4 = 1$

110 → 6^{th} bit has error.